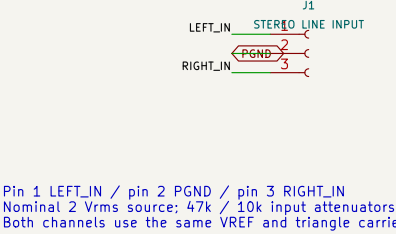


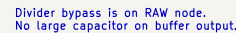
STEREO LINE INPUT



Vout = 1.185 x (1 + Rtop/Rbottom)
11.9922 V nominal; divider current 118.5 uA
EN tied to IN; DNC pin 7 left open; EP to PGND

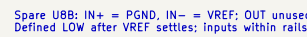
$V_{out} = 1.185 \times (1 + R_{top}/R_{bottom})$
 5.0007 V nominal; divider current 118.5 μ A
 EN tied to IN; DNC pin 7 left open; EP to PGND

3. SHARED 400 kHz OSCILLATOR



Positive-feedback Schmitt: triangle $\rightarrow$ 10k $\rightarrow$ IN+; OUT $\rightarrow$ 49k9 $\rightarrow$ IN+; IN- = VREF
Inverting integrator: square $\rightarrow$ 3k09 $\rightarrow$ IN-; 1nF from OUT to IN-; IN+ = VREF
Ideal 453.722 kHz; 1.9993 to 3.0014 V at nominal 5.0007 V rail. Tune / measure.

DESIGN / COMMISSIONING NOTES



Open-loop Rev 1: no audio feedback or protection.
Target 30 W/channel into 8 ohms at +24 V.
Power target and timing require bench validation.

Input attenuation = $10k / (47k + 10k) = 0.17544$.
AC corner = $1 / (2 \pi \times 57k \times 220n) = 12.69 \text{ Hz}$.
RF slope approx. 193 kHz from $(47k \parallel 10k) / 100p$.
2 Vrms input gives 0.351 Vrms / 0.496 Vpk modulation

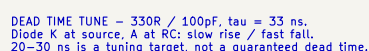
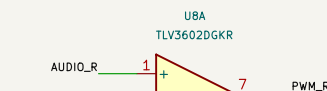
Non-Inverting Schmitt + Inverting Integrator:
 $V_{tri} = V_{ref} + (10k / 499k) \times (V_{ref} - V_{QSG})$
 $f = 499k / (4 \times 10k \times 30k \times 1n) = 403,722 \text{ kHz}$
 Comparator delay, hysteresis and op-amp dynamics
 change actual amplitude / period. Measure and tune.
 Dead time is provisional: RC tau = 33 ns.
 Ideal rise to 2.5 V takes 22.9 ns at a 5 V source.
 Threshold spread, propagation skew, diode current,
 parasitics, driver and MOSFET switching all matter.
 Measure all four gates before selecting final timing.

Both channels default DISABLED (10k pulldowns).
DNP 1k pullups permit later standalone enabling.

PGND is the single global board-wide 0 V reference. Speaker outputs float; do not ground either terminal. Pull capacitors are shared; label after unique value.

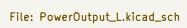
Metric SMD R/C land patterns assigned, no placement.
Film/filter capacitor MPNs and thermals need checking
before PCB layout / manufacture.

5. RIGHT INPUT + PWM + DEAD TIME



DEAD TIME TUNE = $330R / 100\mu F$, $\tau_{\text{au}} = 33$ ns.
Diode K at source, A at RC; slow rise / fast fall.
20-30 ns is a tuning target, not a guaranteed dead time.

DEAD TIME TUNE = 330R / 100pF, tau = 33 ns.
Diode K at source, A at RC; slow rise / fast fall.
20-30 ns is a tuning target, not a guaranteed dead time.



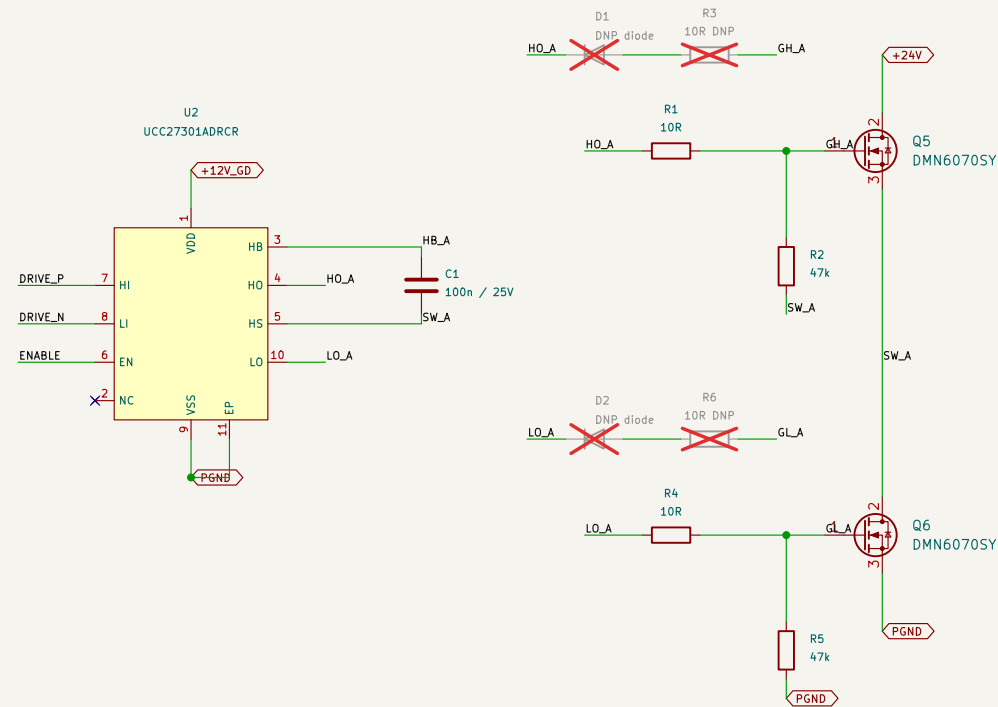
File: PowerOutput_R_kload_sch

Enable defaults LOW. Fit DNP 1k only for standalone enable.
100n here belongs to U4; Schmitt package bypass is in IC POWER row.

Enable defaults LOW. Fit DNP 1k only for standalone enable.
100n here belongs to U9; Schmitt package bypass is in IC POWER row.

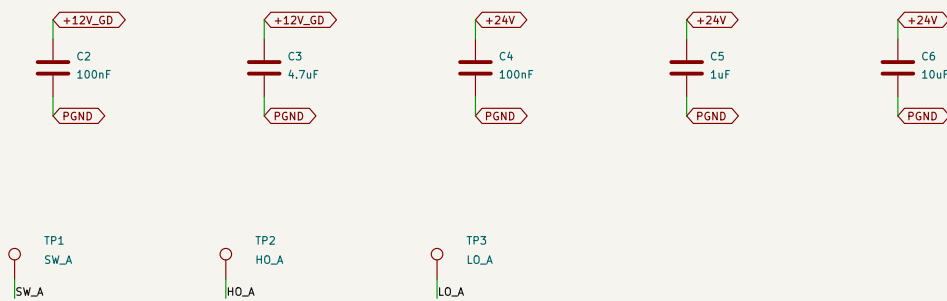
INCOMPLETE DRAFT – ENGINEERING DECISIONS PENDING
OPA1656 supply unassigned: +5 V common-mode range is insufficient.
Saved DMN6070SY retained provisionally; AGENTS.md specifies DMTH6016PS.

HALF BRIDGE A



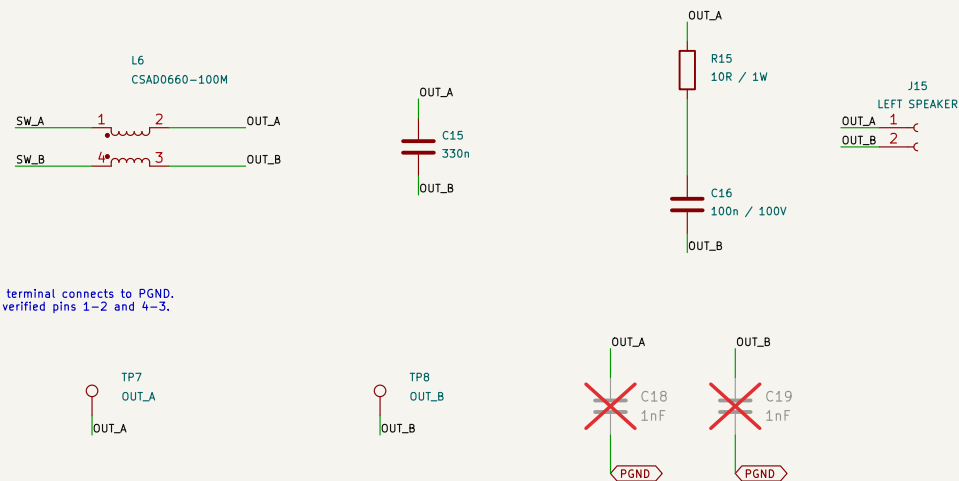
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics



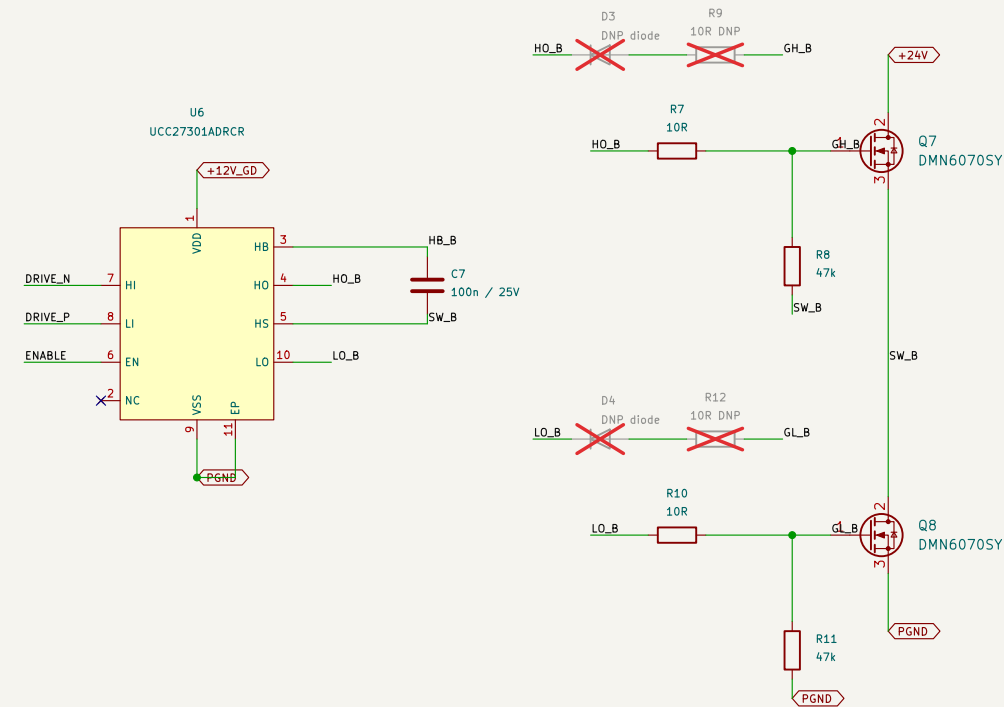
HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time.
DNP diode cathode faces driver: optional faster turn-off path.

DIFFERENTIAL OUTPUT / 8 OHM SPEAKER



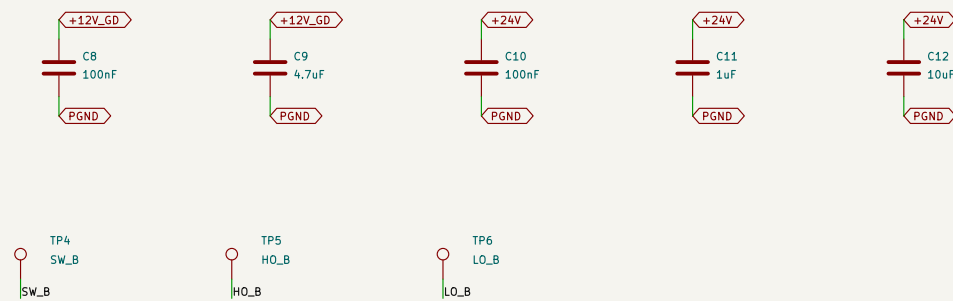
Speaker floats: neither terminal connects to PGND.
L6: 10uH per winding; verified pins 1-2 and 4-3.

HALF BRIDGE B



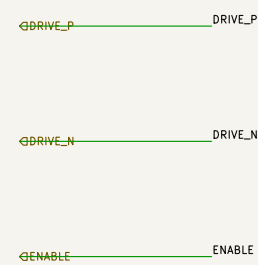
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics

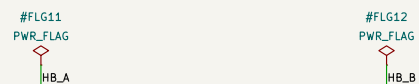


HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time.
DNP diode cathode faces driver: optional faster turn-off path.

CHANNEL CONTROL / SHARED SUPPLIES



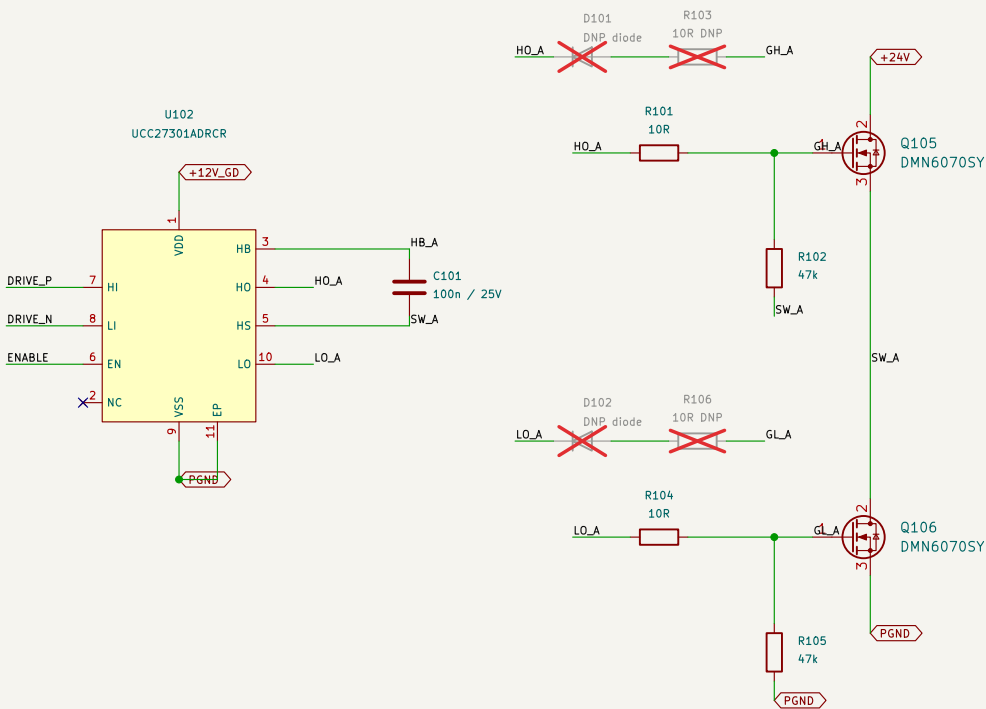
Bridge A: HI = DRIVE_P, LI = DRIVE_N.
Bridge B: HI = DRIVE_N, LI = DRIVE_P.
ENABLE drives both EN pins. Defaults LOW on main.
+26V / +12V_GD / PGND are global.
Speaker and switch nets remain local to this channel.
PGND is the single board-wide 0 V reference.



HB flags represent charging through each UCC27301A internal bootstrap diode.

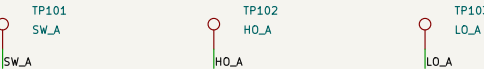
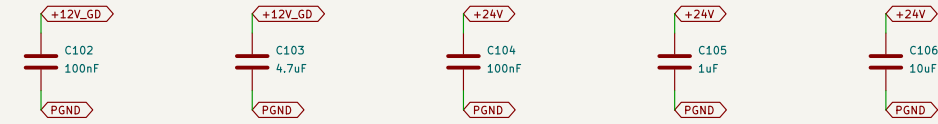
Sheet: /LEFT OUTPUT/ File: PowerOutput_L.kicad_sch	
Title: LEFT BTL power output	
Size: A3	Date:
KiCad E.D.A. 10.05	Rev: Stereo Rev 1 Id: 2/3

HALF BRIDGE A



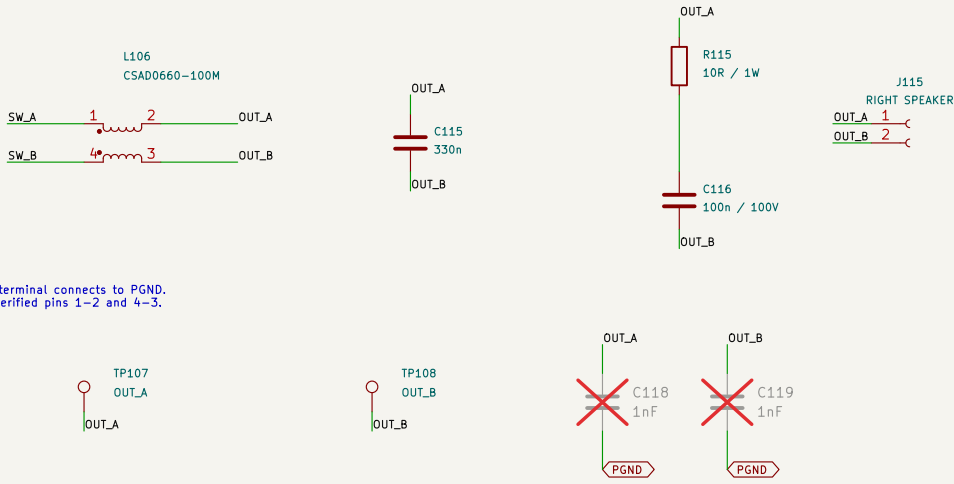
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics



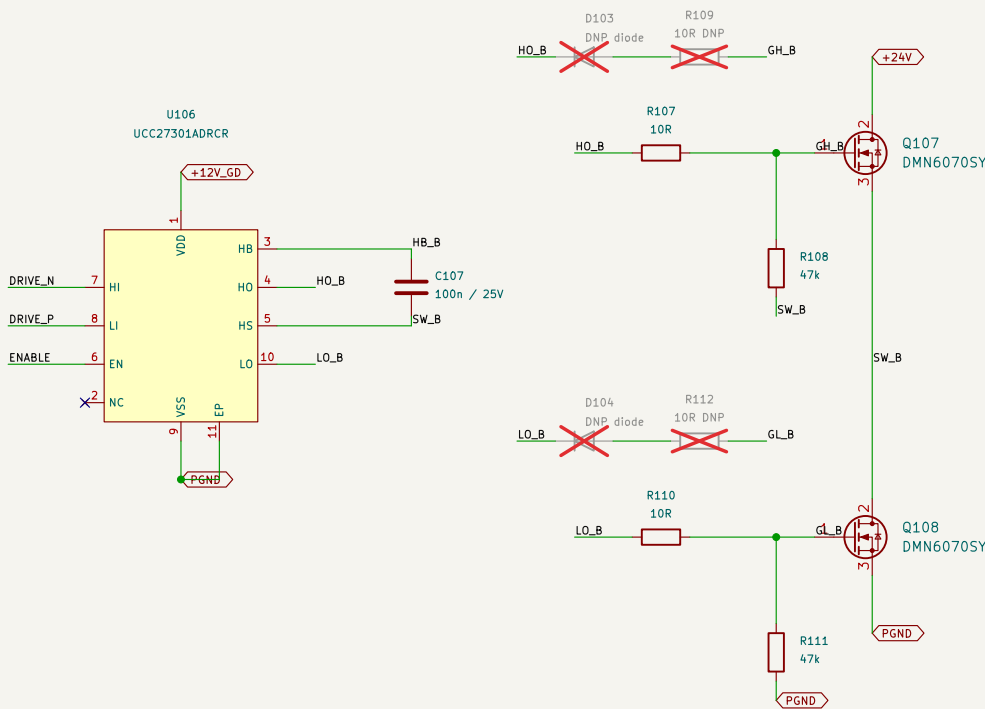
HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time.
DNP diode cathode faces driver: optional faster turn-off path.

DIFFERENTIAL OUTPUT / 8 OHM SPEAKER



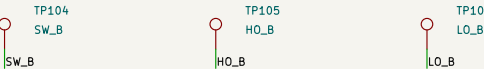
Speaker floats: neither terminal connects to PGND.
L6: 10uH per winding; verified pins 1-2 and 4-3.

HALF BRIDGE B



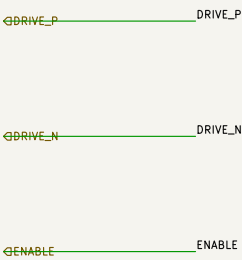
Driver bypass: at VDD/VSS

Local 24 V power loop ceramics



HS / VSS: Kelvin returns to respective MOSFET sources.
Complementary DRIVE_P / DRIVE_N include external dead time.
DNP diode cathode faces driver: optional faster turn-off path.

CHANNEL CONTROL / SHARED SUPPLIES



Bridge A: HI = DRIVE_P, LI = DRIVE_N.
Bridge B: HI = DRIVE_N, LI = DRIVE_P.
ENABLE drives both EN pins. Defaults LOW on main.
+24V / +12V_GD / PGND are global.
Speaker and switch nets remain local to this channel.
PGND is the single board-wide 0 V reference.

HB flags represent charging through each UCC27301A internal bootstrap diode.

Sheet: /RIGHT OUTPUT/ File: PowerOutput_R.kicad_sch		
Title: RIGHT BTL power output		
Size: A3	Date:	Rev: Stereo Rev 1
KiCad E.D.A. 10.0.5		Id: 3/3